

MNIST Handwritten Digit Recognition Report

Objective

The objective of this project is to develop a machine learning model capable of recognizing handwritten digits using the MNIST dataset.

Dataset Description

The MNIST dataset contains 70,000 grayscale images representing handwritten digits from 0 to 9.

Each image:

- Has dimensions of 28×28 pixels
- Contains 784 pixel features after flattening

Target classes:

- Digits 0–9

Data Preprocessing

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

A `random_state` value of 42 was used to ensure reproducible results.

Feature Scaling

`StandardScaler` was applied to normalize feature values.

Benefits:

- Faster convergence
- More stable optimization
- Improved model training efficiency

Model Selection

A Logistic Regression classifier was selected because it is a simple and effective baseline model for multi-class classification problems.

Model Parameters:

- max_iter = 1000

Model Evaluation

The model was evaluated using:

- Training Accuracy
- Testing Accuracy

Training Accuracy indicates how well the model learned from the training dataset.

Testing Accuracy measures the model's ability to generalize to unseen data.

Results

Training Accuracy: 0.94525

Testing Accuracy: 0.9155

Prediction Analysis

Correctly classified digit images were visualized to verify successful predictions.

Incorrectly classified digit images were analyzed to identify challenging samples where handwritten digits were ambiguous or poorly written.

Observations

- Logistic Regression achieved high classification accuracy on the MNIST dataset.
- Most digit images were classified correctly.
- Misclassifications generally occurred between visually similar digits.
- Feature scaling improved training stability.

Conclusion

The Logistic Regression model successfully classified handwritten digits with high accuracy. The project demonstrates a complete machine learning workflow including data preprocessing, model training, evaluation, and prediction analysis.